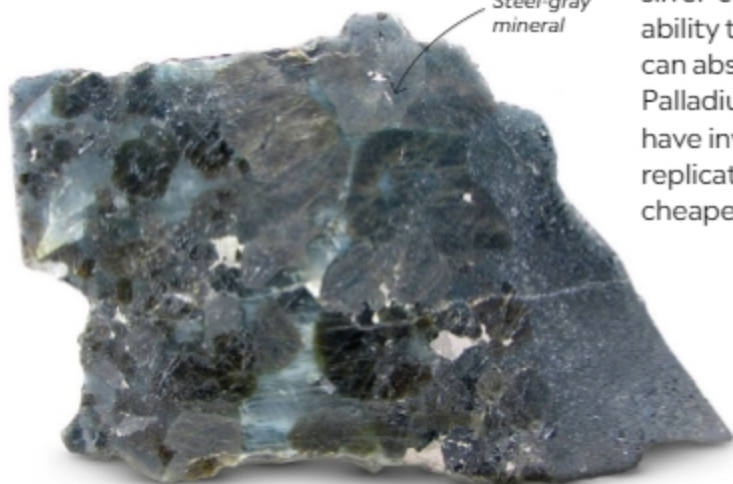


Palladium

46 Pd This precious metal is far rarer than silver or gold. It is used extensively in jewelry, particularly in “white gold,” where gold is mixed with other silver-colored metals. A curious property of palladium is its ability to “soak up” hydrogen gas: at room temperature, it can absorb up to 900 times its own volume of hydrogen. Palladium is too expensive to use this way, but scientists have investigated the possibility of replicating this property in a cheaper material.

Pure palladium bar refined in a laboratory

Steel-gray mineral



Braggite

Although palladium is found pure in nature, it does also appear in some rare minerals, such as braggite. Discovered in 1932, this mineral also contains platinum, nickel, and sulfur.



Catalytic converter

Along with rhodium, palladium is used to coat the insides of catalytic converters for cars. Harmful gases, such as carbon monoxide, pass through the catalytic converter, where the palladium helps them react so they are converted into carbon dioxide, water, and nitrogen. These are then expelled by the car's exhaust systems.

Silver

47 Ag This precious metal has been used for centuries, both as a form of currency and to craft jewelry. As the compound silver nitrate, it has been used as an antibiotic and a disinfectant. Some silver compounds, such as silver bromide, are light-sensitive; they change color when exposed to light. This property made the development of black-and-white photography possible.



Pellet of pure silver

Crystal-like structure

Chlorargyrite

One of the main ores of silver, chlorargyrite is chiefly composed of the compound silver chloride. It is colorless until it is exposed to light, when it turns brown and purple.



Using silver

Due to its shiny appearance, silver has been used in jewelry and in decorative items, from candlesticks to cutlery. One drawback of silver is that when exposed to air, it tarnishes easily to produce black-colored silver sulfide. As a result, decorative items that contain silver need to be cleaned regularly.

